

Q 96 $a^x = b$, $b^y = c$, $c^z = a$, find $xyz = ?$ (given $a, b, c \neq 1, -1$)

Q 96 $(x+1)^{x+4} = 1$, find x .

Soln

$a^b = 1$ $\begin{cases} \rightarrow b=0, a \text{ is non zero} \\ \rightarrow a=1 \\ \rightarrow a=-1, b=\text{even} \end{cases}$

$$(x+1)^{x+4} = 1$$

$$x+4=0$$

$$x = -4$$

$$(-3)^0 = 1 \quad \checkmark$$

$$x+1=1$$

$$x=0$$

$$x+1=-1$$

$$x=-2$$

$$(-1)^2 = 1 \quad \checkmark$$

solution $x = -4, 0, -2$

given $a^x = b$, $b^y = c$, $c^z = a$

$\rightarrow a^x = b$

$(c^z)^x = b \quad (a = c^z)$

$c^{zx} = b$

$(b^y)^{zx} = b \quad (c = b^y)$

$b^{xyz} = b$

$xyz = 1$

Ratio $\rightarrow$ To compare things

$\rightarrow$ units should be same

$\rightarrow \frac{a}{b} \Rightarrow a:b$
 $\uparrow$ antecedent $\rightarrow$ consequent

$\rightarrow a:b = c:d$
 $\Rightarrow \frac{a}{b} = \frac{c}{d}$

$\rightarrow 2:3 \Rightarrow 4:6$

$\rightarrow$ lesser inequality

$a < b$

greater inequality

$a > b$

Q 96 $(x+y):(x+2y) = 5:8$, find ratio of $\frac{2x+3y}{3x+2y} = ?$

Q 96 $a:b = 2:3$ and $b:c = 4:5$, find $a:b:c = ?$

Q 96 $6x^2 - 5xy + y^2 = 0$, then $x:y = ?$

Q 96 2 numbers are in the ratio $3:4$ and their LCM is 180
 find those numbers. (also their HCF)

Soln $(x+y):(x+2y) = 5:8$

$$\Rightarrow \frac{x+y}{x+2y} = \frac{5}{8}$$

$$\Rightarrow 8(x+y) = 5(x+2y)$$

$$\Rightarrow 8x + 8y = 5x + 10y$$

$$\Rightarrow 3x = 2y$$

$$\Rightarrow x = \frac{2}{3}y$$

$$\frac{2x+3y}{3x+2y} = \frac{2\left(\frac{2}{3}y\right) + 3y}{3\left(\frac{2}{3}y\right) + 2y}$$

$$= \frac{\frac{4}{3}y + 3y}{2y + 2y}$$

$$= \frac{\cancel{y}\left(\frac{4}{3} + 3\right)}{4y}$$

$$= \frac{\frac{13}{3}}{4}$$

$$= \frac{13}{12}$$

Soln2

$$a:b = 2:3 \text{ and } b:c = 4:5$$
$$= 8:12 \quad | \quad = 12:15$$

$$a:b:c = 8:12:15$$

Soln $6x^2 - 5xy + y^2 = 0$, $x:y$

$$6\frac{x^2}{y^2} - \frac{5xy}{y^2} + \frac{y^2}{y^2} = 0$$

$$6\left(\frac{x}{y}\right)^2 - 5\left(\frac{x}{y}\right) + 1 = 0$$

$$\frac{x}{y} = t, \quad 6t^2 - 5t + 1 = 0$$

$$6t^2 - 3t - 2t + 1 = 0$$

$$3t(2t-1) - 1(2t-1) = 0$$

$$(2t-1)(3t-1) = 0$$

$$2t-1=0 \quad | \quad 3t-1=0$$

$$t = \frac{1}{2} \quad | \quad t = \frac{1}{3}$$

$$\frac{x}{y} = \frac{1}{2} \quad | \quad \frac{x}{y} = \frac{1}{3}$$

Soln numbers are $3x$ and $4x$

$$\text{LCM}(3x, 4x) = 12x$$

A/Q $12x = 180$

$$x = 15$$

$$\begin{array}{l|l} \text{numbers} = 3x & 4x \\ = 45 & = 60 \end{array}$$

For two numbers

$$\text{LCM} \times \text{HCF} = \text{Product of numbers}$$

$$180 \times \text{HCF} = 45 \times 60$$

$$\text{HCF} = \frac{15 \times 60}{180} = 15$$

Proportion (equation)

ie $\frac{a}{b} = \frac{c}{d} \Leftrightarrow a:b = c:d$, then a, b, c, d are in proportion
 $\rightarrow d$ is the 4th proportion of a, b, c



Properties

① $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$, ie $\rightarrow$ product of means = product of extremes.

② Continued proportion

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} = \dots$$

special case

$$\frac{a}{b} = \frac{b}{c}$$

→ b is called mean proportion of a and c ($b > 0$)

→ c is called third proportion of a and b .

Q find third proportion of 12 and 18.

$$a = 12, b = 18, c = ?$$

$$\frac{12}{18} = \frac{18}{c}$$

$$\Rightarrow 2c = 54$$

$$\Rightarrow c = 27$$

③ $\frac{a}{b} = \frac{c}{d}$
 $\frac{b}{a} = \frac{d}{c}$ (invertendo)

④ $\frac{a}{b} = \frac{c}{d}$
 $\Rightarrow \frac{a}{c} = \frac{b}{d}$ [alternando]

a, d can also be interchanged

⑤ componendo/dividendo.

9/ $\frac{a}{b} = \frac{c}{d}$

then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

eg $a^2 + b^2 = 117$, $ab = 54$, $(a > b > 0)$ $\frac{a+b}{a-b} = ?$

$$\frac{a^2 + b^2}{ab} = \frac{117}{54}$$

$$\frac{a^2 + b^2}{2ab} = \frac{117}{108}$$

$$\frac{a^2 + b^2 + 2ab}{a^2 + b^2 - 2ab} = \frac{117 + 108}{117 - 108}$$

$$\frac{(a+b)^2}{(a-b)^2} = 25$$

$$\frac{a+b}{a-b} = \pm 5$$

$$\frac{a+b}{a-b} = 5$$

Q Find x

$$\frac{x^2 - x + 1}{x^2 + x + 1} = \frac{2x - 1}{2x + 1}$$

$$\Rightarrow \frac{(x^2 - x + 1) + (x^2 + x + 1)}{(x^2 - x + 1) - (x^2 + x + 1)} = \frac{(2x - 1) + (2x + 1)}{(2x - 1) - (2x + 1)}$$

$$\Rightarrow \frac{\cancel{x^2} - \cancel{x} + 1 + \cancel{x^2} + \cancel{x} + 1}{\cancel{x^2} - \cancel{x} + 1 - \cancel{x^2} - \cancel{x} - 1} = \frac{\cancel{2x} - 1 + \cancel{2x} + 1}{\cancel{2x} - 1 - \cancel{2x} - 1}$$

$$\Rightarrow \frac{2x^2 + 2}{-2} = \frac{4x}{-2}$$

$$\Rightarrow \frac{\cancel{2}(x^2 + 1)}{x} = \frac{\cancel{2}}{4x}$$

$$\Rightarrow (x^2 + 1) = 2x^2$$

$$x^2 + 1 = 2x^2$$

$$x^2 = 1$$

$$\therefore x = \pm 1$$

Q $\sqrt{a^2 + b^2} \sqrt{x^2 + y^2} = ax + by$

then show that

$$\frac{a}{x} = \frac{b}{y}$$

$$\sqrt{a^2+b^2} \sqrt{x^2+y^2} = ax+by$$

$$(a^2+b^2)(x^2+y^2) = (ax+by)^2 \quad (sbs)$$

$$\Rightarrow \cancel{a^2x^2} + a^2y^2 + b^2x^2 + \cancel{b^2y^2} = \cancel{a^2x^2} + \cancel{b^2y^2} + 2axby$$

$$\Rightarrow a^2y^2 + b^2x^2 - 2axby = 0$$

$$\Rightarrow (ay - bx)^2 = 0$$

$$\Rightarrow ay - bx = 0$$

$$\Rightarrow ay = bx$$

$$\Rightarrow \frac{a}{x} = \frac{b}{y}$$

set theory

Sets: A collection of well defined objects or elements

A = set of natural numbers less than 5 ✓ →

$$A = \{4, 2, 3, 1\}$$

B = set of tall students in 18th may batch ✗

C = set of students in 18th may batch who are atleast 170 cm tall. ✓

D = list of good Prime minister in India ✗

Properties

- enclosed by curly brackets
- elements are separated by comma
- order of elements does not matter
- no repetition of elements
- generally denoted by capital letters

Cardinality $\rightarrow$ no of elements in a set.

$$A = \{a, b, c, 3\}, \quad n(A) = 4$$

$$B = \{\underline{1}, \underline{2}, \underline{(3 \& 4)}, \underline{\phi}, \underline{\text{horse}}\}, \quad n(B) = 5$$

$$C = \text{set of natural number}, \quad n(C) = \infty$$

$$D = \phi \quad \left| \quad D = \text{set of natural numbers less than } (-2), \quad n(D) = 0 \right.$$

$$D = \{ \}$$

Null set or empty set:

If $n(A) = 0$, then A is null set.

Null set is represent by ϕ or $\{ \}$

Representation of sets

$A =$ natural numbers less than 6

Roster form

$$A = \{1, 2, 3, 4, 5\}$$

set builder form

$$\{x : \text{a property of } x\}$$

↓ such that
include x

$$A = \{x : x \text{ is a natural number and } x < 6\}$$

$$A = \{2x \mid x^2 = 1\}$$

$$\therefore A = \{2, -2\}$$

Q Write the following in Roster form.

$$a) A = \{x+1 \mid x^2 - 5x + 6 = 0\}$$

$$b) B = \{(a, b) \mid a \text{ is a natural number less than 4 and } b = a^2\}$$

Soln a) $A = \{3, 4\}$

$$\begin{aligned}x^2 - 5x + 6 &= 0 \\x^2 - 2x - 3x + 6 &= 0 \\x(x-2) - 3(x-2) &= 0 \\(x-2)(x-3) &= 0 \\x &= 2, x = 3 \\x+1 &= 3, 4\end{aligned}$$

b) $B = \{(a, b) \mid a \text{ is a natural number less than 4 and } b = a^2\}$

$B = \{(1, 1), (2, 4), (3, 9)\}$

$a=1$	$a=2$	$a=3$
$b=1$	$b=4$	$b=9$
$(1, 1)$	$(2, 4)$	$(3, 9)$

Notation

$\in$ $\rightarrow$ belongs to

$\cup$ $\rightarrow$ union

$-$ $\rightarrow$ set difference

$\notin$ $\rightarrow$ does not belong to

$\cap$ $\rightarrow$ intersection

$\subseteq$ $\rightarrow$ subset

$\subset$ $\rightarrow$ proper subset

$\forall$ $\rightarrow$ for all

$\exists$ $\rightarrow$ there exists

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eg: $A = \{a, b, c, 4\}$ $a \in A$ $5 \notin A$

set operations (union, intersection, difference)

$$\underline{A} = \{1, 2, 3, p, q\} \quad B = \{2, 3, 4, q, r\} \quad , \quad C = \{3, p, 5\}$$

$$A \cup B = \{1, 2, 3, p, q, 4, r\}, \quad B \cap C = \{3\}, \quad A - B = \{1, p\}$$

$$A \cap (B \cup C) =$$

$$\underline{\underline{B \cup C}} = \{2, 3, 4, 9, p, 5\}$$

$$A \Delta B = (A - B) \cup (B - A)$$

$$= \alpha(1, p) \cup \{4, n\} = \alpha(1, 4, p, n)$$

$A - B = \{1, p\}$
 $\downarrow$
 in A, but not in B

$$B - C = \{2, 4, 9, x\}$$

$$C-B = \{p, 5\}$$

Symmetric difference $\rightarrow$ A, B are two sets then
 $(A-B) \cup (B-A)$ is symmetric difference $(A \Delta B)$

Properties

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

$$A - B \neq B - A$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \left[\text{distributive law} \right]$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\begin{aligned} & B \cup (B \cap C) \\ &= (B \cup B) \cap (B \cup C) \\ &= B \cap (B \cup C) \end{aligned}$$

complement of an event and Universal set

→ for event A , A^c is called complement of A .

A^c → those elements which are not in A .

universal set:

$$U = \{1, 2, 3, 4, 5, 6, 7, a, b, c\}$$

$$A = \{1, 2, 3, 4\}$$

$$A^c = \{5, 6, 7, a, b, c\}$$

$$A^c = U - A$$

$$\rightarrow A \cap A^c = \phi$$

$$\rightarrow A \cup A^c = U$$

subset: A is called subset of B [$A \subseteq B$], if every element of set A belongs to B .

eg $A = \{1, 2, p\}$

$$B = \{1, 2, 3, p, q\}$$

$$C = \{p, 1, 2\}$$

$$D = \{\} \quad , \quad E = \phi$$

$$A \subseteq B$$

$$C \subseteq B$$

$$B \subseteq A$$

$$A \subseteq C$$

$$C \subseteq A$$

$$A \subseteq D$$

$$D \subseteq A$$

$$D \subseteq E$$

→ Null set is subset of every set.

proper subset: A is called proper subset of B ($A \subset B$) if
 $A \subset B$ and there exists an additional element in B.

$$A = \{1, 2, p\}, \quad B = \{1, 2, 3, p, q\}, \quad C = \{p, 1, 2\}, \quad D = \{ \}, \quad E = \phi$$
$$\begin{array}{ccccccc} A \subset B & , & C \subset B & , & A \subset C & , & D \subset A \\ \checkmark & & \checkmark & & \times & & \checkmark \\ & & & & & & D \subset E \\ & & & & & & \times \end{array}$$

→ Empty set is proper subset of every non empty set.

Equal set: A and B are equal sets if they both contain same elements.

→ $A \subset B$ and $B \subset A$ then $A = B$

power set: $P(A)$ is called power set A if it contain every subset of A .

eg: $A = \{1, 2\}$, $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

if number of element in A is ' n '

then no of subsets of $A = 2^n$

no of proper subset of $A = 2^n - 1$

no of elements in power set of $A = 2^n$

Q if no of subsets of set A is 56 more than the no of subsets of B ,
then find no of elements in set A and set B .

$$n(A) = a, \quad n(B) = b$$

$$2^a - 2^b = 56$$

$$2^b \left(\frac{2^a}{2^b} - 1 \right) = 2^3 \times 7$$

$$2^b \left(\underbrace{2^{a-b}}_{\substack{\downarrow \\ \text{even}}} - \underbrace{1}_{\text{odd}} \right) = \underbrace{2^3}_{\text{even}} \times \underbrace{7}_{\text{odd}}$$

$$\begin{array}{l|l} 2^b = 2^3 & 2^{a-b} - 1 = 7 \\ \hline b = 3 & 2^{a-3} = 8 \\ & a-3 = 3 \\ & a = 6 \end{array}$$

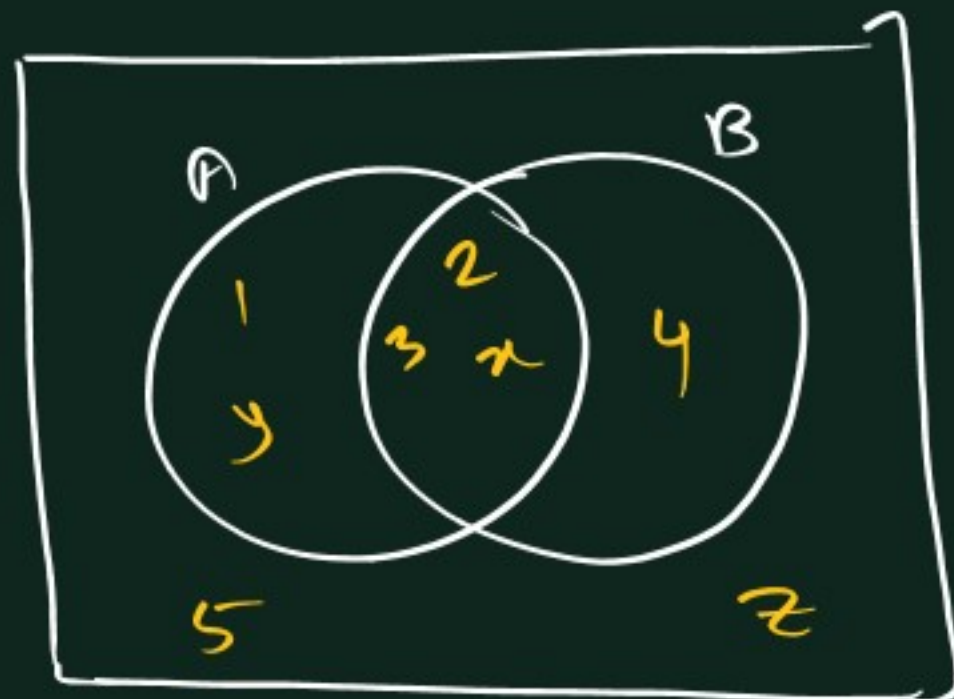
$$\begin{array}{r} 2 \overline{) 56} \\ 2 \overline{) 28} \\ 2 \overline{) 14} \\ 7 \end{array}$$

Venn diagram: pictorial representation of sets

eg $U = \{1, 2, 3, 4, 5, x, y, z\}$

$$A = \{1, 2, 3, x, y\}$$

$$B = \{2, 3, 4, x\}$$



eg: $A \subset B$



eg $A \cap B = \emptyset$



shading regions in venn diagram:

